VELS

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AGRI-GUARDIAN

AgriGuardian
Healthy Plants • Smarter Care

AN AI -POWERED PLANT DISEASE DIAGNOSIS WITH DECISION SUPPORT SYSTEM

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PROJECT GUIDE

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WHAT IS AGRI-GUARDIAN

- Agri Guardian is an AI-powered plant-care application designed for beginners.
- It helps users choose suitable plants, identify plants and diseases, assess disease severity, and understand plant-care needs.
- The app provides guidance on watering, sunlight, soil, treatment, organic/chemical options, and prevention.
- It also offers AI assistance, reminders, and plant growth monitoring.
- Overall, Agri Guardian acts as a personal AI gardening companion for healthy plant growth.

PROBLEM STATEMENT

- Beginners often lack knowledge about plant selection and proper care.
- Identifying plant diseases and their severity can be difficult.
- Existing apps often provide only individual plant-related services.
- Users need simple guidance for treatment and prevention.
- Agri Guardian aims to provide an AI-based solution for plant selection, diagnosis, care, and health monitoring.





OBJECTIVES OF AGRI-GUARDIAN

- RECOMMEND SUITABLE PLANTS BASED ON THE USER'S ENVIRONMENT AND PREFERENCES.
- IDENTIFY PLANTS AND DETECT POSSIBLE DISEASES USING AI AND COMPUTER VISION.
- ASSESS DISEASE SEVERITY BY ANALYZING AFFECTED AREAS OF THE PLANT. PROVIDE PERSONALIZED GUIDANCE FOR WATERING, SUNLIGHT, SOIL,
- TREATMENT, AND PREVENTION.
- PROVIDE AN AI-BASED PLANT ASSISTANT AND TRACK PLANT HEALTH AND GROWTH OVER TIME.

LITERATURE REVIEW

1. Plant Disease Detection

CNN-based disease classification.
Gap: No complete plant-care support.

2. PlantDoc

Real-world plant disease dataset.
Gap: Limited coverage.

3. EfficientNet

Accurate, efficient image classification.

Gap: Not plant-care specific.

LITERATURE REVIEW

4. MobileViT
Lightweight mobile vision model.
Gap: Needs task-specific training.

5. U-Net
Segments affected image regions.
Gap: Needs adaptation for plants.

Research Gap
Existing systems focus on individual tasks. Agri Guardian combines diagnosis, severity, treatment, prevention, and plant monitoring in one app.

PROPOSED METHODOLOGY

1. Knowledge Retrieval – Use RAG to retrieve relevant agricultural information.
2. Recommendations – Generate treatment, organic, chemical-control, and prevention guidance.
3. AI Assistance & Monitoring – Provide personalized guidance and track plant health over time.
4. Severity Analysis – Detect affected areas using U-Net/YOLO-Seg.
5. Knowledge Retrieval – Use RAG to retrieve relevant agricultural information.

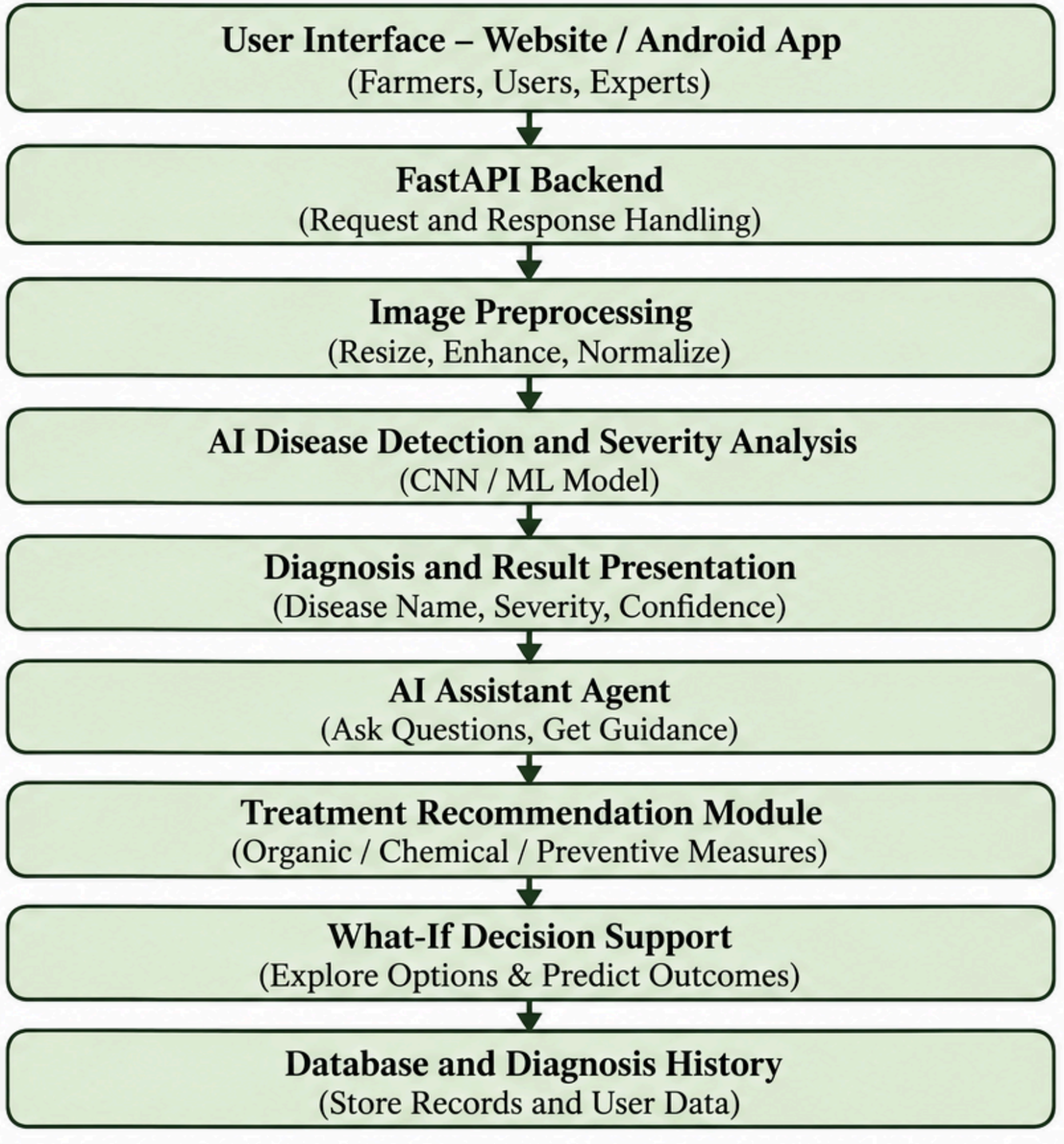
KEY MODULES

1. User & Plant Profile – Stores user preferences and plant details.
2. Plant Recommendation – Suggests suitable plants based on environment.
3. Plant Identification – Identifies plants from images.
4. Disease Diagnosis – Detects possible plant diseases using AI.

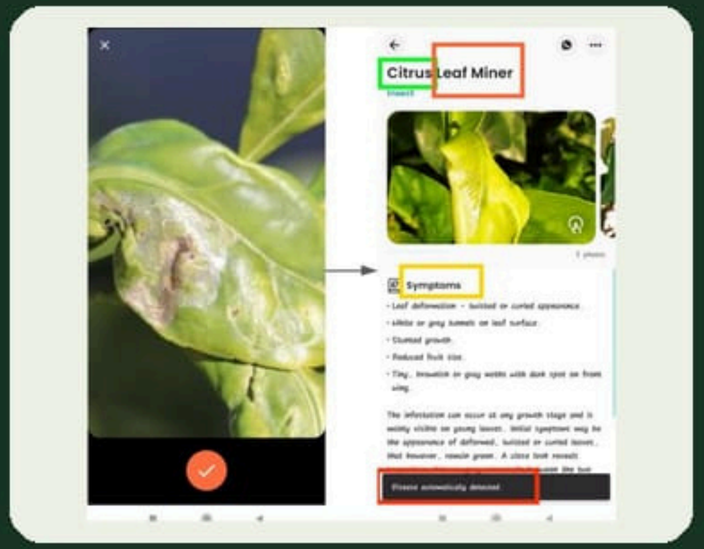
KEY MODULES

5. Severity Analysis – Estimates the affected area and severity.
6. Treatment & Prevention – Provides organic, chemical-control, and prevention information.
7. AI Assistant – Answers plant-care questions.
8. Growth Monitoring – Tracks plant health and growth over time.

AgriGuardian – System Architecture

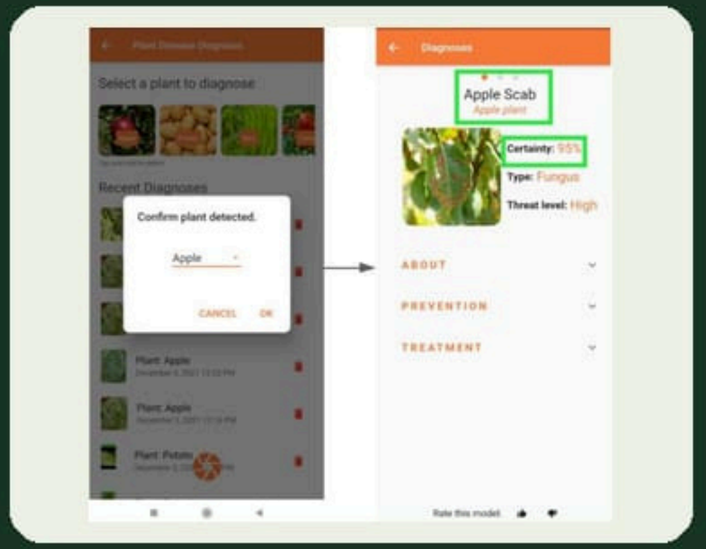


AI-powered plant-care applications



AGROAI-PLANT

Disease identification, treatment suggestions and crop-health monitoring.



AGROAI-PLANT

Also discusses weather forecasts and fertilizer guidance.



LEAF DOCTOR

Assesses disease severity and highlights affected leaf regions.

Image analysis, sensors and training

IMAGE DATA

Collect photographs of healthy and diseased plants, label the examples and train a model to recognize symptoms.

IOT SENSOR DATA

Temperature, humidity and soil-moisture readings can help identify conditions associated with disease risk.

EXPERT SYSTEMS

Diagnostic systems can combine observed symptoms with stored knowledge to support treatment decisions.

MOBILE ACCESS

Smartphone applications can let users photograph a plant and receive a diagnosis directly.

AI for plant disease detection

The reviewed study explores how artificial intelligence can help farmers identify plant diseases quickly and make better crop-management decisions.

WHY IT MATTERS

Plant diseases can reduce crop yield and cause financial losses. Early identification gives farmers more time to take preventive or corrective action.

WHAT AI ADDS

Machine learning and deep learning analyze plant images and other field data to recognize disease patterns, estimate severity and support diagnosis.

Source: Judijanto et al. (2024), Journal of Infrastructure, Policy and Development, 8(13), 6920.

Conclusion: smarter plant health management

Artificial intelligence can strengthen plant-disease management by combining image recognition, prediction, monitoring and decision support.

MAIN TAKEAWAY

AI can help farmers identify problems earlier and respond with more informed preventive or corrective actions.

RESPONSIBLE ADOPTION

Reliable model training, data security, privacy and practical accessibility are essential for long-term use.

Reference: Judijanto, L. et al. (2024). “Revolutionizing plant health with machine learning for disease detection and diagnosis.” Journal of Infrastructure, Policy and Development, 8(13), 6920. https://doi.org/10.24294/jipd6920

Detecting disease from plant images



1 CAPTURE

A farmer or plant enthusiast takes a photograph of a leaf or plant showing possible symptoms.

2 ANALYZE

Computer vision and deep-learning models examine visible patterns and return a possible disease diagnosis.

The paper describes PlantifyDr as an example of an app that analyzes an uploaded plant image and provides a diagnosis.

Benefits and implementation challenges

POTENTIAL BENEFITS

• Faster disease identification • Earlier warning and intervention • More informed treatment decisions • Reduced crop losses and improved productivity • More efficient use of farm resources

KEY CHALLENGES

• Reliable AI systems require investment and training • Models depend on suitable, well-labelled data • Accuracy and accessibility need further improvement • Farmer data security and privacy must be protected

The article emphasizes that careful management is needed for sustainable adoption.

Research methods and data



IMAGE-BASED ANALYSIS

Photographs of affected leaves are processed to learn visual symptoms and disease patterns.

MACHINE LEARNING

Models are trained with labelled examples so they can classify plant conditions and recognize recurring features.

The article discusses AI, machine learning, deep learning, image processing and connected sensor data as approaches used in plant-health research.

From diagnosis to decision support

PREDICT

AI can use weather patterns, crop varieties and population-related factors to estimate the risk of disease spread.

RECOMMEND

After diagnosis, systems can suggest suitable treatment or preventive steps based on the detected condition.

MONITOR

Regular monitoring and early alerts help farmers notice changes and act before symptoms spread further.

The study presents these capabilities as ways to reduce crop losses and support more timely farm decisions.

AgriGuardian

Smarter Farming • Healthier Crops • A Sustainable Future

RESULTS

1 Image Analysis & Disease Detection

The uploaded plant image was analyzed using AI image processing techniques. The system successfully detected the affected crop and identified the disease with high accuracy (92%). This helps farmers quickly recognize issues in their crops without manual inspection.

2 Disease Diagnosis Result

The system identified the disease as Early Blight (Tomato). It provided a confidence score of 92% and showed the affected crop as Tomato. The result also displayed the symptoms matching the detected disease from the database.

3 Severity and Risk Assessment

The disease severity is classified as High Risk. The system calculated the spread risk as High and predicted a severe yield impact. This assessment helps farmers take immediate action to prevent further damage and reduce crop loss.

4 Recommendations and Prevention

The system suggests using recommended fungicide (e.g., Mancozeb 75% WP – 2g/L) and advises removing infected leaves to stop the spread. It also recommends maintaining proper irrigation and avoiding water stagnation, keeping proper plant spacing, and monitoring the field regularly.